

# The Human–AI Proxy Pre-Singularity

Mathematical, Physical, and Environmental Constraints as Catalysts of Emergent Intelligence

*A preprint-level operational theory, diagnostics, and test protocols*

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## Abstract

We develop an operational theory of the *pre-singularity* as a measurable regime of coupled dynamics among humans, artificial systems, and the physical environment—where the environment becomes computationally active rather than merely a passive constraint. The central thesis is that “singularity-like” transitions are not best characterized as isolated model capability thresholds, but as *boundary conditions in environmental agency*: sensing, memory, inference, and optimization become embedded into space (edge inference, actuation fabrics, place-based memory, and energy/compute routing) such that the environment itself participates in cognition and begins to architect human trajectories by shaping the action-feasibility set. We formalize this shift with (i) a proxy framework for coupled objectives, (ii) constraint-manifold dynamics where the manifold evolves under embedded computation, (iii) environmental coherence via spectral connectivity of a dynamic environment graph, and (iv) phase-transition criteria using order parameters tied to loop latency, observability, and actuation authority. We propose falsifiable hypotheses, a composite *Pre-Singularity Index* (PSI) designed to resist “vibe-based” inflation, and a set of causal identification protocols (randomized interventions, difference-in-differences, interrupted time series, and instrument-based designs) to distinguish correlation from genuine environment-to-human causal influence. Finally, we specify a Minimal Viable Field (MVF) architecture and scalable experimental plans (room → building → campus) with governance invariants (auditability, real override, anti-monopolization) and privacy-by-design requirements. This manuscript is intended to be preprint-rigorous: mechanism-first, measurable, and explicitly falsifiable.

## Contents

|          |                                                                             |          |
|----------|-----------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                         | <b>4</b> |
| 1.1      | Motivation: from tools to environmental co-authorship . . . . .             | 4        |
| 1.2      | Contributions (explicit) . . . . .                                          | 4        |
| 1.3      | Scope and non-claims . . . . .                                              | 4        |
| <b>2</b> | <b>Background and Related Work (Conceptual + Technical)</b>                 | <b>5</b> |
| 2.1      | Cybernetics, feedback, and requisite variety . . . . .                      | 5        |
| 2.2      | Distributed cognition and the extended mind . . . . .                       | 5        |
| 2.3      | Ubiquitous computing, context-aware systems, ambient intelligence . . . . . | 5        |
| 2.4      | Edge, fog, cloudlets, and compute placement . . . . .                       | 5        |
| 2.5      | Network science and spectral structure . . . . .                            | 5        |

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| 2.6       | Safety-critical control: MPC and barrier functions . . . . .       | 5         |
| 2.7       | Causal inference and intervention-based science . . . . .          | 5         |
| <b>3</b>  | <b>Notation, System Model, and Assumptions</b>                     | <b>6</b>  |
| 3.1       | State spaces . . . . .                                             | 6         |
| 3.2       | Control inputs and observations . . . . .                          | 6         |
| 3.3       | Coupled dynamics (continuous-time) . . . . .                       | 6         |
| 3.4       | Constraints as a manifold . . . . .                                | 6         |
| 3.5       | Assumptions (minimal, explicit) . . . . .                          | 7         |
| <b>4</b>  | <b>The Proxy Framework</b>                                         | <b>7</b>  |
| 4.1       | Proxy definition (operational) . . . . .                           | 7         |
| 4.2       | Human–AI proxy effect (triadic) . . . . .                          | 7         |
| 4.3       | Environmental agency (measurable) . . . . .                        | 7         |
| 4.4       | The sub-environment as the boundary layer . . . . .                | 8         |
| <b>5</b>  | <b>Environmental Coherence via Graph and Spectral Structure</b>    | <b>8</b>  |
| 5.1       | Environment graph . . . . .                                        | 8         |
| 5.2       | Edge weight schema (engineering-friendly) . . . . .                | 8         |
| 5.3       | Laplacian and coherence . . . . .                                  | 8         |
| 5.4       | Coherence supports coordination (formal exemplar) . . . . .        | 8         |
| 5.5       | Coherence is not enough: anti-monopoly constraints . . . . .       | 9         |
| <b>6</b>  | <b>Field Emergence as a Phase Transition</b>                       | <b>9</b>  |
| 6.1       | Loop latency and the attention boundary . . . . .                  | 9         |
| 6.2       | Order parameter . . . . .                                          | 9         |
| 6.3       | Interpretation . . . . .                                           | 9         |
| <b>7</b>  | <b>Information, Observability, and the Cost of Authority</b>       | <b>10</b> |
| 7.1       | Latent state estimation . . . . .                                  | 10        |
| 7.2       | Mutual information (or proxy) as a metric . . . . .                | 10        |
| <b>8</b>  | <b>Constraint-Manifold Dynamics and “Environment as Architect”</b> | <b>10</b> |
| 8.1       | Affordance operator . . . . .                                      | 10        |
| 8.2       | Constraint shaping index . . . . .                                 | 10        |
| <b>9</b>  | <b>Safety-Critical Environmental Actuation</b>                     | <b>10</b> |
| 9.1       | Bounded actuation and rate limits . . . . .                        | 10        |
| 9.2       | Safe set invariance (CBF viewpoint) . . . . .                      | 11        |
| 9.3       | Projection filter (engineering-friendly) . . . . .                 | 11        |
| 9.4       | Human override as a constraint (not a UI) . . . . .                | 11        |
| <b>10</b> | <b>From Postulate to Diagnosis: Falsifiable Hypotheses</b>         | <b>11</b> |
| <b>11</b> | <b>The Pre-Singularity Index (PSI): A Conservative Diagnostic</b>  | <b>12</b> |
| 11.1      | Component indices (all normalized to $[0, 1]$ ) . . . . .          | 12        |
| 11.2      | Composite PSI . . . . .                                            | 13        |

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>12 Causal Tests and Identification (Credibility Upgrade)</b> | <b>14</b> |
| 12.1 Randomized interventions (gold standard) . . . . .         | 14        |
| 12.2 Quasi-experiments . . . . .                                | 14        |
| 12.3 Causal dominance score . . . . .                           | 14        |
| 12.4 Causal ADI(strong form) . . . . .                          | 14        |
| 12.5 Ethical constraints on experimentation . . . . .           | 14        |
| <b>13 Minimal Viable Field (MVF) Architecture</b>               | <b>15</b> |
| 13.1 Layered stack . . . . .                                    | 15        |
| 13.2 MVF definition . . . . .                                   | 15        |
| 13.3 Closed-loop cognition diagram . . . . .                    | 15        |
| <b>14 Experimental Protocols (Room → Building → Campus)</b>     | <b>15</b> |
| 14.1 Room-scale MVF . . . . .                                   | 15        |
| 14.2 Building-scale MVF . . . . .                               | 16        |
| 14.3 Campus-scale MVF . . . . .                                 | 16        |
| 14.4 “Are we there yet?” checklist (skeptic-proof) . . . . .    | 16        |
| <b>15 Implementation Sketches (Metrics + Coherence + PSI)</b>   | <b>16</b> |
| 15.1 Computing coherence in practice . . . . .                  | 16        |
| 15.2 PSIcomputation . . . . .                                   | 17        |
| <b>16 Governance, Privacy, and Security</b>                     | <b>17</b> |
| 16.1 Governance as constraints . . . . .                        | 17        |
| 16.2 Audit logging . . . . .                                    | 17        |
| 16.3 Threat model (minimum viable) . . . . .                    | 18        |
| 16.4 Privacy-by-design . . . . .                                | 18        |
| 16.5 Goodhart and runaway proxy optimization . . . . .          | 18        |
| <b>17 Discussion, Limitations, and Predictions</b>              | <b>18</b> |
| 17.1 What would count as evidence? . . . . .                    | 18        |
| 17.2 Limitations . . . . .                                      | 18        |
| 17.3 Predictions (testable) . . . . .                           | 19        |
| <b>18 Conclusion</b>                                            | <b>19</b> |
| <b>A Appendix A: Formal Field Intelligence Functionals</b>      | <b>20</b> |
| <b>B Appendix B: Minimal Data Schema for Measuring PSI</b>      | <b>20</b> |
| <b>C Appendix C: Field Layers Diagram Template</b>              | <b>21</b> |

# 1 Introduction

## 1.1 Motivation: from tools to environmental co-authorship

The dominant public framing of “singularity” focuses on model capability in isolation: a system surpasses a human benchmark, then surpasses all benchmarks, then recursively self-improves. That framing is incomplete as an engineering theory of *real-world transitions*. The decisive boundary, we argue, is not merely *how smart* a device or model is, but *where cognition lives* and *how it couples* to physical reality.

When computation is sparse and external (apps, devices, dashboards), humans remain primary architects of the environment. When computation becomes embedded into the environment itself—dense sensing, persistent place-memory, low-latency inference, and real actuation that reshapes affordances—the environment begins to shape human trajectories continuously. At that point, the environment is no longer a passive stage; it becomes a *closed-loop participant*. This is the *human-AI proxy pre-singularity*: a regime where (i) AI increasingly proxies human planning and inference, (ii) humans proxy AI deployment and value constraints, and (iii) the environment proxies both by shaping constraints, defaults, and routes.

## 1.2 Contributions (explicit)

This preprint contributes:

- C1. Mechanism-first formalization:** a coupled dynamical model  $(\mathbf{H}, \mathbf{A}, \mathbf{E})$  where *the environment is stateful and actuating*, not merely a disturbance.
- C2. Operational definitions:** environmental agency, sub-environment boundary layer, and “field-dominant” regime via order parameters.
- C3. Spectral coherence grounding:** an environment graph  $G_E(t)$  with coherence  $\kappa_E(t) = \lambda_2(L_E(t))$  as a tractable integration metric, plus anti-monopoly constraints.
- C4. Falsifiable hypotheses and diagnostics:** a conservative PSIindex requiring coherence, loop compression, observability, actuation, place-memory, energy robustness, and causal dominance.
- C5. Causal inference upgrade:** explicit identification strategies (randomized interventions, DiD, ITS, IV) to test environment→human effects.
- C6. Prototype pathway:** Minimal Viable Field (MVF) architecture and experimental protocols spanning room/building/campus scales, including fault injection and red-team evaluation.
- C7. Governance and safety invariants:** auditable control, real override, bounded actuation, privacy-by-design, and threat modeling for adversarial sensing/actuation.

## 1.3 Scope and non-claims

We do *not* claim a metaphysical singularity, nor that any single metric can certify a world-historical phase transition. We claim that: (i) there exists a rigorous, testable notion of a *pre-singularity regime* defined by environmental agency and causal dominance, and (ii) this notion can be measured and falsified within bounded regions (rooms/buildings/campuses), enabling credible engineering and scientific evaluation.

## 2 Background and Related Work (Conceptual + Technical)

This section positions the theory in established literature to avoid category errors (philosophy treated as engineering, or engineering treated as mysticism).

### 2.1 Cybernetics, feedback, and requisite variety

Classical cybernetics formalizes intelligence as feedback: sensing, control, adaptation, and stability under disturbance. This manuscript is fundamentally cybernetic: the environment becomes a controller with internal state, and pre-singularity corresponds to a regime shift in loop dominance [1, 2].

### 2.2 Distributed cognition and the extended mind

The claim that cognition can span artifacts and environment is well-developed in distributed cognition and extended mind theory [3, 4]. Our contribution is to treat this as an empirical systems hypothesis with measurable loop dynamics, rather than a purely philosophical stance.

### 2.3 Ubiquitous computing, context-aware systems, ambient intelligence

Ubiquitous computing (“calm technology”) anticipates computation diffusing into everyday environments [5]. Context-aware computing and ambient intelligence provide the design lineage for sensing + inference embedded into places [6, 7]. “Smart environments” literature offers practical architecture precedents [8].

### 2.4 Edge, fog, cloudlets, and compute placement

When latency matters, cloud-centric architectures shift to edge/fog/cloudlets for local inference and actuation [9, 10]. The core transition studied here is not only edge inference, but *edge inference plus persistent place-memory plus actuation authority*, yielding a closed-loop “field.”

### 2.5 Network science and spectral structure

Spectral quantities of graphs (e.g., Laplacian eigenvalues) characterize connectivity, mixing, and convergence in distributed processes [11, 12]. We leverage algebraic connectivity  $\lambda_2$  as a tractable coherence measure for an embedded environment substrate.

### 2.6 Safety-critical control: MPC and barrier functions

If the environment actuates, it must be safe and stable under constraints. Model Predictive Control (MPC) provides a standard constraint-handling paradigm [13, 14]. Control Barrier Functions (CBFs) provide set invariance tools for enforcing safety properties under real-time optimization [15, 16].

### 2.7 Causal inference and intervention-based science

Pre-singularity is a claim about causal dominance, not correlation. Modern causal inference formalizes intervention, identifiability, and robust evaluation methods [18, 19, 20, 21].

### 3 Notation, System Model, and Assumptions

#### 3.1 State spaces

We model three coupled subsystems:

$$\mathbf{H}(t) \in \mathbb{R}^{n_H} \quad (\text{human cognitive/behavioral state}), \quad (1)$$

$$\mathbf{A}(t) \in \mathbb{R}^{n_A} \quad (\text{artificial system/model state}), \quad (2)$$

$$\mathbf{E}(t) \in \mathbb{R}^{n_E} \quad (\text{environmental computational field state}). \quad (3)$$

The *environment* is not merely geometry:

$$\mathbf{E}(t) \equiv (\mathbf{E}_{\text{sense}}(t), \mathbf{E}_{\text{infer}}(t), \mathbf{E}_{\text{act}}(t), \mathbf{E}_{\text{mem}}(t), \mathbf{E}_{\text{energy}}(t)). \quad (4)$$

#### 3.2 Control inputs and observations

We introduce control inputs:

$$u_H(t) \in \mathbb{R}^{m_H} \quad (\text{human actions}), \quad (5)$$

$$u_A(t) \in \mathbb{R}^{m_A} \quad (\text{AI outputs / policies}), \quad (6)$$

$$u_E(t) \in \mathbb{R}^{m_E} \quad (\text{environmental actuation controls}). \quad (7)$$

Observations are:

$$y_H(t) = O_H(\mathbf{H}, \mathbf{E}, t) + \eta_H(t), \quad (8)$$

$$y_A(t) = O_A(\mathbf{A}, \mathbf{E}, t) + \eta_A(t), \quad (9)$$

$$y_E(t) = O_E(\mathbf{E}, \mathbf{H}, \mathbf{A}, t) + \eta_E(t), \quad (10)$$

where  $\eta$ . represent noise/disturbance and partial observability.

#### 3.3 Coupled dynamics (continuous-time)

We define coupled dynamics:

$$\frac{d}{dt}\mathbf{H} = f(\mathbf{H}, \mathbf{A}, \mathbf{E}, u_H, u_A, u_E, t), \quad (11)$$

$$\frac{d}{dt}\mathbf{A} = g(\mathbf{A}, \mathbf{H}, \mathbf{E}, u_H, u_A, u_E, t), \quad (12)$$

$$\frac{d}{dt}\mathbf{E} = h(\mathbf{E}, \mathbf{H}, \mathbf{A}, u_H, u_A, u_E, t). \quad (13)$$

#### 3.4 Constraints as a manifold

Reality imposes constraints; admissible states lie on a (possibly time-varying) constraint manifold:

$$x(t) := (\mathbf{H}(t), \mathbf{A}(t), \mathbf{E}(t)) \in \mathcal{M}(t) \subseteq \mathbb{R}^{n_H+n_A+n_E} \quad (14)$$

with constraint functions:

$$c_i(x(t), t) \leq 0 \quad (\text{energy, latency, safety, physical feasibility, policy}). \quad (15)$$

**Remark 1.** *A key move: in field regimes,  $\mathcal{M}(t)$  is not static. Embedded computation can rewrite effective constraints (routes, access, friction, availability), hence the constraint surface itself becomes partially learned and rewritten.*

### 3.5 Assumptions (minimal, explicit)

**Assumption 1** (Bounded actuation and sampling). *Environmental actuation is bounded in magnitude and rate:*

$$\|u_E(t)\|_2 \leq U_{\max}, \quad \left\| \frac{d}{dt} u_E(t) \right\|_2 \leq R_{\max}, \quad (16)$$

and the environment loop is sampled at finite rate with measurable latency  $\tau(t)$ .

**Assumption 2** (Observability sufficient for authority). *If  $\mathbf{E}$  has strong actuation authority over a variable, it maintains sufficient observability of the relevant latent state (formalized in §7 and §12).*

**Assumption 3** (Audit and override exist). *The environment control channel includes an auditable log and a real override mechanism with bounded time-to-effect (formalized in §16).*

## 4 The Proxy Framework

### 4.1 Proxy definition (operational)

Let  $Y$  be a subsystem with objective functional  $J_Y$  over horizon  $T$ :

$$J_Y = \int_0^T r_Y(x(t), u(t), t) dt. \quad (17)$$

**Definition 1** (Proxy). *We say subsystem  $X$  is a proxy for  $Y$  over horizon  $T$  if the coupling induced by  $X$  increases achievable performance for  $Y$  under comparable constraints:*

$$\sup_{u \in \mathcal{U}} J_Y(\text{coupled to } X) > \sup_{u \in \mathcal{U}} J_Y(\text{decoupled from } X). \quad (18)$$

### 4.2 Human–AI proxy effect (triadic)

In the pre-singularity regime:

- **AI  $\rightarrow$  human proxy:** AI acts as proxy for search, planning, compression, inference, and coordination.
- **Human  $\rightarrow$  AI proxy:** humans proxy deployment, value constraints, boundary setting, data/actuation placement.
- **Environment  $\rightarrow$  both proxy:** environment proxies cognition via affordance shaping, routing, nudging, and place-memory.

### 4.3 Environmental agency (measurable)

**Definition 2** (Environmental agency). *Environmental agency measures responsiveness of  $\mathbf{E}$  to human/AI state and inputs:*

$$\text{EAI}(t) \triangleq \left\| \frac{\partial h}{\partial \mathbf{H}} \right\|_2 + \left\| \frac{\partial h}{\partial \mathbf{A}} \right\|_2 + \left\| \frac{\partial h}{\partial u_H} \right\|_2 + \left\| \frac{\partial h}{\partial u_A} \right\|_2. \quad (19)$$

Passive environment:  $\text{EAI} \approx 0$ . Pre-singularity:  $\text{EAI}$  rises and becomes trajectory-dominant.

#### 4.4 The sub-environment as the boundary layer

**Definition 3** (Sub-environment). *The sub-environment is the instrumentation + inference + actuation layer mediating lived experience:*

$$\mathbf{E}_{sub} = \mathbf{E}_{sense} \oplus \mathbf{E}_{infer} \oplus \mathbf{E}_{act} \oplus \mathbf{E}_{mem} \oplus \mathbf{E}_{energy}. \quad (20)$$

**Remark 2.** *This is the boundary between “tools in space” and “space as a system.” The theory claims that singularity-like transitions occur when  $\mathbf{E}_{sub}$  becomes dense/coherent/autonomous enough to function as a field-scale optimizer.*

### 5 Environmental Coherence via Graph and Spectral Structure

#### 5.1 Environment graph

We represent the embedded environment as a dynamic weighted graph:

$$G_E(t) = (V(t), E(t), W(t)), \quad (21)$$

where each node is a micro-site (sensor/edge compute/actuator/memory/energy router), and each edge weight reflects effective coupling quality.

#### 5.2 Edge weight schema (engineering-friendly)

A practical model:

$$w_{ij}(t) = \alpha \text{BW}_{ij}(t) - \beta \text{LAT}_{ij}(t) - \gamma \text{ENERGY}_{ij}(t) - \delta \text{RISK}_{ij}(t), \quad (22)$$

with  $\alpha, \beta, \gamma, \delta \geq 0$  chosen per domain.

#### 5.3 Laplacian and coherence

Let  $W$  be symmetric weighted adjacency and  $D = \text{diag}(W\mathbf{1})$ . Laplacian:

$$L_E(t) = D(t) - W(t). \quad (23)$$

Let  $\lambda_2(t)$  be the second-smallest eigenvalue of  $L_E(t)$ .

**Definition 4** (Environmental coherence). *Define:*

$$\kappa_E(t) \triangleq \lambda_2(L_E(t)). \quad (24)$$

**Remark 3.**  $\kappa_E$  is a tractable measure of whether the environment behaves like a coherent substrate (field) or a fragmented set of components.

#### 5.4 Coherence supports coordination (formal exemplar)

Consider continuous-time consensus dynamics  $\dot{z} = -L_E z$  on a connected graph.

**Theorem 1** (Consensus rate lower bounded by  $\lambda_2$ ). *If  $G_E$  is connected and  $L_E$  is the (weighted) Laplacian, then for  $z(t)$  evolving under  $\dot{z} = -L_E z$ , the disagreement energy decays at a rate governed by  $\lambda_2$ :*

$$\|z(t) - \bar{z}\mathbf{1}\|_2 \leq e^{-\lambda_2 t} \|z(0) - \bar{z}\mathbf{1}\|_2, \quad (25)$$

where  $\bar{z} = \frac{1}{n}\mathbf{1}^\top z(0)$ .

*Proof sketch.*  $L_E$  is symmetric positive semidefinite with eigenvalues  $0 = \lambda_1 < \lambda_2 \leq \dots$ . Projecting onto the orthogonal complement of  $\mathbf{1}$  yields exponential decay at rate at least  $\lambda_2$ . See standard spectral graph/consensus results [11, 12].  $\square$

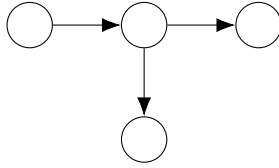
### 5.5 Coherence is not enough: anti-monopoly constraints

High coherence can be achieved by either (i) a healthy mesh, or (ii) centralized dominance. Field viability requires coherence *and* decentralization.

**Definition 5** (Anti-monopolization constraints). *Let weighted degree be  $d_i = \sum_j w_{ij}$ . Enforce:*

$$\max_i d_i(t) \leq d_{\max} \quad \text{and optionally} \quad \frac{\max_i d_i(t)}{\frac{1}{n} \sum_i d_i(t)} \leq \rho_{\max}, \quad (26)$$

*to limit hub domination under fixed coherence targets.*



Coherence can come from  
mesh health *or*  
central dominance.  
Field safety requires  
coherence + anti-hub bounds.

## 6 Field Emergence as a Phase Transition

### 6.1 Loop latency and the attention boundary

Define end-to-end loop latency:

$$\tau(t) = \tau_{\text{sense}}(t) + \tau_{\text{infer}}(t) + \tau_{\text{act}}(t) + \tau_{\text{prop}}(t). \quad (27)$$

The core mechanism is *loop compression*: when  $\tau$  falls beneath relevant human operational timescales (attention reorientation, task-switching), the environment becomes continuously steering rather than occasionally advising.

### 6.2 Order parameter

We define an order parameter capturing field emergence:

$$\Phi(t) \triangleq \sigma\left(a \kappa_E(t) - b \tau(t) + c \mathcal{O}(t) + d \mathcal{U}(t)\right) \in [0, 1], \quad (28)$$

where  $\mathcal{O}$  measures observability quality and  $\mathcal{U}$  measures actuation authority;  $\sigma(\cdot)$  is a squashing function.

**Definition 6** (Field-dominant regime). *The system is field-dominant if  $\Phi(t) \geq \Phi_*$  over sustained macroscopic windows (hours/days), not merely transient spikes.*

### 6.3 Interpretation

Below threshold: embedded systems assist, but humans remain primary architects. Above threshold: the environment begins to architect human trajectories systematically by shaping constraints and defaults continuously.

## 7 Information, Observability, and the Cost of Authority

### 7.1 Latent state estimation

Let  $s(t)$  be a latent world state relevant to tasks (positions, flows, hazards, intent proxies, resource levels). Define an environment estimator:

$$\hat{s}_E(t) = \mathcal{F}(y_E(0:t), u_E(0:t)). \quad (29)$$

### 7.2 Mutual information (or proxy) as a metric

Define:

$$\mathcal{I}_E(t) \triangleq I(s(t); \hat{s}_E(t)). \quad (30)$$

In practice, mutual information can be approximated by prediction skill, calibrated likelihoods, or task-relevant estimator error:

$$\mathcal{L}_E(t) := \mathbb{E}[\|s(t) - \hat{s}_E(t)\|_2^2]. \quad (31)$$

**Proposition 1** (Authority without observability increases instability risk). *If actuation authority is strong while  $\mathcal{I}_E$  is low (or  $\mathcal{L}_E$  is high), then environmental control is more likely to introduce harmful steering via misestimated state. Thus field emergence requires observability commensurate with authority.*

## 8 Constraint-Manifold Dynamics and “Environment as Architect”

### 8.1 Affordance operator

Define an affordance operator  $\mathbf{A}$  mapping environmental state to a human action-feasibility set:

$$\mathbf{A} : \mathbf{E} \mapsto \mathcal{U}_H(\mathbf{E}, t) \subseteq \mathbb{R}^{m_H}. \quad (32)$$

The environment “architects” humans by modifying  $\mathcal{U}_H(\mathbf{E}, t)$  continuously.

### 8.2 Constraint shaping index

We define constraint shaping strength as sensitivity of feasible action sets to environment state:

$$\text{CSI}(t) \triangleq \left\| \frac{\partial \mathcal{U}_H(\mathbf{E}, t)}{\partial \mathbf{E}} \right\|_2, \quad (33)$$

where  $\frac{\partial \mathcal{U}_H}{\partial \mathbf{E}}$  is interpreted via set-derivative surrogates (e.g., changes in feasible region volume, boundary displacement under small perturbations, or changes in default policy constraints).

**Remark 4.** *This is the technical bridge from philosophical statement to measurable mechanism: the environment becomes architect of the human  $\iff$  the environment increasingly determines feasible/default actions.*

## 9 Safety-Critical Environmental Actuation

### 9.1 Bounded actuation and rate limits

Restating Assumption 1, environmental actuation must be bounded to prevent runaway steering:

$$\|u_E(t)\|_2 \leq U_{\max}, \quad \left| \frac{d}{dt} u_E(t) \right| \leq R_{\max}. \quad (34)$$

## 9.2 Safe set invariance (CBF viewpoint)

Let  $\mathcal{S} = \{x : b(x) \geq 0\}$  be a safe set. Control Barrier Functions enforce forward invariance of  $\mathcal{S}$  by constraining  $u_E$  such that

$$\dot{b}(x) + \alpha(b(x)) \geq 0, \quad (35)$$

for suitable class- $\mathcal{K}$  function  $\alpha$ , yielding real-time constraints inside a quadratic program [15, 16].

## 9.3 Projection filter (engineering-friendly)

A pragmatic safety wrapper is projecting proposed actuation into a safe, bounded set (convex approximation):

Listing 1: Sketch: project proposed actuation into a bounded safe set

```
import numpy as np

def project_to_safe(u_proposed, U_max, rate_max, u_prev, dt):
    u = u_proposed.astype(float).copy()

    # magnitude bound
    nrm = np.linalg.norm(u) + 1e-12
    if nrm > U_max:
        u *= (U_max / nrm)

    # rate bound
    du = (u - u_prev) / max(dt, 1e-9)
    du_nrm = np.linalg.norm(du) + 1e-12
    if du_nrm > rate_max:
        u = u_prev + du * (rate_max / du_nrm) * dt

    return u
```

## 9.4 Human override as a constraint (not a UI)

**Definition 7** (Override guarantee). *There exists an override action channel  $u_H^{override}(t)$  such that for any admissible  $x(t)$ , the system can be driven back into a safe admissible region within bounded time  $T_{ovr}$ :*

$$\forall t, \exists u_H^{override}(t) \text{ s.t. } x(t + T_{ovr}) \in \mathcal{M}_{safe}. \quad (36)$$

This prevents the “override theater” failure mode where override exists only nominally.

# 10 From Postulate to Diagnosis: Falsifiable Hypotheses

We treat “pre-singularity” as an empirical hypothesis about *where cognition lives* and *which subsystem dominates trajectories*.

## H1: Constraint shaping dominance

**Hypothesis H1.** Over time, measurable variation in human outcomes is increasingly explained by variation in  $\mathcal{U}_H(\mathbf{E}, t)$  rather than by static constraints or internal deliberation alone.

**Falsifier.** Controlled changes in  $\mathbf{E}$  (thus  $\mathcal{U}_H$ ) do *not* produce measurable changes in behavior/outcomes.

## H2: Loop compression into attention timescales

**Hypothesis H2.** A significant fraction of daily decisions are influenced by loops with latency below a reference attention timescale  $\tau_{\text{attn}}$ :

$$\Pr[\tau(t) \leq \tau_{\text{attn}}] \text{ increases over time.} \quad (37)$$

**Falsifier.** If  $\tau$  remains slow (or loops rarely affect decisions), H2 fails.

## H3: Field coherence above fragmentation threshold

**Hypothesis H3.** The environment exhibits sustained coherence:

$$\kappa_E(t) \geq \kappa_\star \quad \text{over macroscopic windows.} \quad (38)$$

**Falsifier.**  $\kappa_E$  remains near zero except in isolated subsystems.

## H4: Environmental agency is causal, not correlational

**Hypothesis H4.** Environmental actuation changes human trajectories causally (intervention-sensitive), not merely correlationally. **Falsifier.** Randomized/quasi-random perturbations fail to change outcomes; causal models reject environment  $\rightarrow$  human influence.

# 11 The Pre-Singularity Index (PSI): A Conservative Diagnostic

To detect a pre-singularity regime, we define a composite index designed to resist inflation by mere device usage.

## 11.1 Component indices (all normalized to $[0, 1]$ )

### (1) Environmental Coherence Index ECI

$$\text{ECI}(t) \triangleq \text{clip}_{[0,1]} \left( \frac{\kappa_E(t) - \kappa_{\min}}{\kappa_{\max} - \kappa_{\min}} \right). \quad (39)$$

### (2) Loop Compression Index LCI Let $\tau_{\text{ref}}$ be a domain reference latency:

$$\text{LCI}(t) \triangleq \text{clip}_{[0,1]} \left( 1 - \frac{\tau(t)}{\tau_{\text{ref}}} \right). \quad (40)$$

### (3) Observability Coverage Index OCI Let $\tilde{\mathcal{I}}_E(t)$ be a normalized observability score and $C_{\text{sense}}(t)$ be sensing coverage:

$$\text{OCI}(t) \triangleq \text{clip}_{[0,1]} \left( \omega_I \tilde{\mathcal{I}}_E(t) + (1 - \omega_I) C_{\text{sense}}(t) \right). \quad (41)$$

### (4) Actuation Coverage Index ACI Let $C_{\text{act}}(t)$ measure the fraction of relevant affordances the environment can modulate:

$$\text{ACI}(t) \triangleq C_{\text{act}}(t) \in [0, 1]. \quad (42)$$

**(5) Memory-in-Place Coverage Index MCI** Let  $C_{\text{mem}}(t)$  measure persistence + retrievability of place-based memory:

$$\text{MCI}(t) \triangleq C_{\text{mem}}(t) \in [0, 1]. \quad (43)$$

**(6) Energy Routing/Autonomy Index ERI** Let  $C_{\text{energy}}(t)$  capture autonomy/robustness of energy supply + shedding behavior:

$$\text{ERI}(t) \triangleq C_{\text{energy}}(t) \in [0, 1]. \quad (44)$$

**(7) Autonomy-Dominance Index ADI** This captures whether the environment is steering outcomes by shaping constraints rather than merely offering advice. Let  $Y$  be an outcome (time-to-completion, safety incidents, path choice, cost). Let  $\Delta Y$  be the intervention effect when environment actuation is toggled/ varied:

$$\text{ADI}(t) \triangleq \text{clip}_{[0,1]} \left( \frac{|\Delta Y(t)|}{|Y_{\text{baseline}}| + \varepsilon} \right). \quad (45)$$

A stronger causal version is defined in §12.

**(8) Constraint Shaping Index CSI**

$$\text{CSI}(t) \triangleq \text{clip}_{[0,1]} \left( \frac{\text{measured constraint-set displacement}}{\text{reference displacement}} \right). \quad (46)$$

(Implemented via set-volume/decision-boundary displacement proxies.)

**(9) Human Retention/Override Index HRI** Measures whether humans can function without the field and whether override is real:

$$\text{HRI}(t) \triangleq \text{clip}_{[0,1]} \left( \underbrace{1 - \Delta_{\text{outage}}}_{\text{independence}} \cdot \underbrace{\exp(-T_{\text{ovr}}/T_0)}_{\text{override speed}} \right). \quad (47)$$

Here  $\Delta_{\text{outage}}$  measures degradation during controlled field outages.

## 11.2 Composite PSI

Let weights be nonnegative and sum to 1:

$$\text{PSI}(t) \triangleq \sum_{k \in \{c, \ell, o, a, m, e, d, s, h\}} w_k \cdot I_k(t), \quad (48)$$

where  $(I_c, I_\ell, I_o, I_a, I_m, I_e, I_d, I_s, I_h) = (\text{ECI}, \text{LCI}, \text{OCI}, \text{ACI}, \text{MCI}, \text{ERI}, \text{ADI}, \text{CSI}, \text{HRI})$ .

**Definition 8** (Operational pre-singularity regime). *A system is in an operational pre-singularity regime if*

$$\text{PSI}(t) \geq \text{PSI}_\star \quad (49)$$

*persistently over a macroscopic window (days to months), with verified causal dominance (§12) and governance invariants (§16).*

**Remark 5.** *This is intentionally hard to satisfy. That hardness is the credibility feature: it prevents “phones + apps” from counting as pre-singularity by default.*

## 12 Causal Tests and Identification (Credibility Upgrade)

Pre-singularity is a claim about causal dominance of environment loops.

### 12.1 Randomized interventions (gold standard)

Randomize an environmental variable  $Z$  (cue intensity, route suggestion, friction parameter) and measure outcome  $Y$ :

$$\text{ATE} = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]. \quad (50)$$

A sustained nontrivial ATE supports H4.

### 12.2 Quasi-experiments

When randomization is infeasible:

- **Difference-in-differences (DiD)**: compare treated vs control zones pre/post deployment.
- **Interrupted time series (ITS)**: detect change points aligned with interventions.
- **Instrumental variables (IV)**: use exogenous variation (e.g., weather-driven occupancy) to identify causal effects.

### 12.3 Causal dominance score

Define:

$$D_{\mathbf{E} \rightarrow H} \triangleq \frac{\text{Var}(\mathbb{E}[Y \mid \text{do}(\mathbf{E})])}{\text{Var}(Y) + \varepsilon}. \quad (51)$$

This is interpretable as the fraction of outcome variance attributable to interventions on environment state.

### 12.4 Causal ADI(strong form)

We can define:

$$\text{ADI}_{\text{causal}} \triangleq \text{clip}_{[0,1]} \left( \frac{D_{\mathbf{E} \rightarrow H}}{D_{\star}} \right), \quad (52)$$

where  $D_{\star}$  is a skeptical reference threshold chosen per domain.

### 12.5 Ethical constraints on experimentation

Environmental interventions can manipulate choices. All experiments must satisfy:

- bounded risk and bounded friction changes,
- opt-out and real override,
- privacy-preserving measurement,
- transparent logging of interventions.

## 13 Minimal Viable Field (MVF) Architecture

### 13.1 Layered stack

A practical field stack:

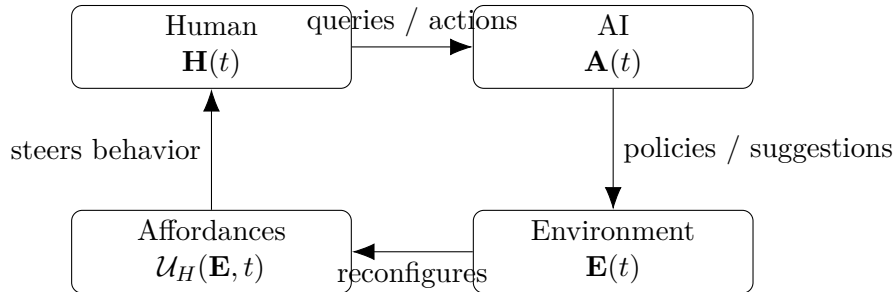
- (1) **Sensing:** multimodal observation (privacy-preserving where possible).
- (2) **Edge inference:** local state estimation, anomaly detection, forecasting.
- (3) **Coordination bus:** low-latency messaging + time sync (NTP/PTP as feasible).
- (4) **Place memory:** event logs, embeddings, maps, and retrieval.
- (5) **Actuation:** physical routing (lighting, signage, access, acoustics, HVAC, robotics).
- (6) **Energy layer:** harvesting/storage/load shedding + power-aware scheduling.

### 13.2 MVF definition

**Definition 9** (Minimal Viable Field). *An MVF is a bounded physical region with:*

- *continuous sensing coverage for task-relevant variables,*
- *local inference with bounded latency,*
- *at least one actuation modality that changes affordances,*
- *place memory that persists across sessions,*
- *measurable coherence  $\kappa_E(t)$  above baseline,*
- *auditable control and real override.*

### 13.3 Closed-loop cognition diagram



## 14 Experimental Protocols (Room $\rightarrow$ Building $\rightarrow$ Campus)

### 14.1 Room-scale MVF

- Define tasks  $\Pi$  (navigation, resource location, hazard alerts, flow stabilization).
- Deploy 2–5 edge nodes + sensors; implement at least one actuation channel.
- Log end-to-end latency  $\tau(t)$ , compute  $\kappa_E(t)$  snapshots, and record outcomes  $Y(t)$ .
- Run controlled interventions  $Z(t)$  for causal effect estimation (bounded changes).

## 14.2 Building-scale MVF

- Introduce zones and sharding: regions  $\{R_k\}$  with local coherence  $\kappa_k$ .
- Add distributed place memory per zone + cross-zone retrieval policies.
- Perform failure injection: drop 20% nodes; measure degradation and recovery.
- Enforce anti-monopoly constraints and measure concentration metrics.

## 14.3 Campus-scale MVF

- Multi-stakeholder objectives: safety, throughput, fairness, energy constraints.
- Governance protocols: audit completeness, override speed distributions.
- Adversarial testing: sensor spoofing, communication degradation, actuation abuse attempts.

## 14.4 “Are we there yet?” checklist (skeptic-proof)

| Criterion           | Measure                                     | What convinces a skeptic?                                   |
|---------------------|---------------------------------------------|-------------------------------------------------------------|
| Coherence           | $\kappa_E = \lambda_2(L_E)$ ; fragmentation | Sustained nontrivial $\kappa_E$ across subsystems           |
| Loop compression    | $\tau$ end-to-end                           | Closed loops faster than operational/attention thresholds   |
| Observability       | $\tilde{\mathcal{I}}_E$ , estimator error   | Stable state estimation under noise/perturbation            |
| Actuation authority | $C_{\text{act}}$ , bounds                   | Environment changes defaults/constraints, not just suggests |
| Place-memory        | $C_{\text{mem}}$ , retrieval success        | Place context persists without re-entry                     |
| Energy autonomy     | $C_{\text{energy}}$ , outage resilience     | Graceful degradation under power/network failures           |
| Causal dominance    | ATE/DiD/ITS, $D_{\mathbf{E} \rightarrow H}$ | Replicable intervention effects on outcomes                 |
| Governance          | override time, audit completeness           | Overrides work; logs explain actions; invariants enforced   |

Table 1: Pre-singularity credibility checklist: measurable and falsifiable.

# 15 Implementation Sketches (Metrics + Coherence + PSI)

## 15.1 Computing coherence in practice

Listing 2: Sketch: compute environmental coherence via lambda2

```
import numpy as np

def coherence_lambda2(W):
    W = 0.5 * (W + W.T)
    D = np.diag(W.sum(axis=1))
    L = D - W
```

```

evals = np.linalg.eigvalsh(L)
evals.sort()
return float(evals[1]) if len(evals) > 1 else 0.0

```

## 15.2 PSIcomputation

Listing 3: Sketch: compute PSI components from logs

```

def clip01(x):
    return max(0.0, min(1.0, float(x)))

def compute_PSI(kappa, kmin, kmax, tau, tau_ref,
                I_norm, C_sense, C_act, C_mem, C_energy,
                ADI, CSI, HRI, w):
    ECI = clip01((kappa - kmin) / max(1e-9, (kmax - kmin)))
    LCI = clip01(1.0 - (tau / max(1e-9, tau_ref)))
    OCI = clip01(0.6 * I_norm + 0.4 * C_sense)

    ACI = clip01(C_act)
    MCI = clip01(C_mem)
    ERI = clip01(C_energy)
    ADI = clip01(ADI)
    CSI = clip01(CSI)
    HRI = clip01(HRI)

    PSI = (w["c"]*ECI + w["l"]*LCI + w["o"]*OCI +
           w["a"]*ACI + w["m"]*MCI + w["e"]*ERI +
           w["d"]*ADI + w["s"]*CSI + w["h"]*HRI)

    return {"ECI":ECI,"LCI":LCI,"OCI":OCI,"ACI":ACI,"MCI":MCI,
           "ERI":ERI,"ADI":ADI,"CSI":CSI,"HRI":HRI,"PSI":PSI}

```

## 16 Governance, Privacy, and Security

### 16.1 Governance as constraints

Encode governance as admissible-set constraints:

$$u_E(t) \in \mathcal{U}_{\text{safe}}(\cdot) \cap \mathcal{U}_{\text{auditable}}(\cdot) \cap \mathcal{U}_{\text{overridable}}(\cdot) \cap \mathcal{U}_{\text{anti-monopoly}}(\cdot). \quad (53)$$

### 16.2 Audit logging

The field must log:

- estimated state  $\hat{s}_E(t)$  and confidence,
- objective terms used and their weights,
- constraint activations and safety projections,
- final  $u_E(t)$  and predicted effects,

- override events and resolution time,
- outcomes  $Y(t)$  for evaluation.

### 16.3 Threat model (minimum viable)

- **Sensor spoofing/adversarial inputs:** corrupt  $y_E$  to steer actuation.
- **Network degradation:** partition  $G_E$  to destroy coherence.
- **Actuation abuse:** push  $u_E$  toward coercive defaults (Goodhart pressure).
- **Opaque control:** remove explainability and destroy governance legitimacy.

### 16.4 Privacy-by-design

Smart environments create surveillance risk by default; privacy must be engineered:

- **Minimize raw capture:** prefer on-device feature extraction over raw video retention.
- **Aggregate + anonymize:** store counts/flows rather than identities.
- **Differential privacy where appropriate:** apply formal privacy guarantees to released aggregates [22, 23].
- **Enterprise risk framing:** align deployments to recognized privacy frameworks [24].

### 16.5 Goodhart and runaway proxy optimization

If the environment optimizes a proxy, it may distort human life around that metric. Governance must enforce multi-objective bounds and regularly re-audit objectives [17].

## 17 Discussion, Limitations, and Predictions

### 17.1 What would count as evidence?

Evidence for a pre-singularity regime (in this operational sense) is *not* subjective amazement. It is sustained high PSI with:

- replicable causal effects environment→human outcomes,
- coherence without monopolization,
- compressed loops operating continuously,
- real override + audit invariants.

### 17.2 Limitations

- **Metric gaming:** any index can be optimized; governance must prevent Goodhart drift.
- **Context dependence:** thresholds  $PSI_*$  are domain-specific; calibration is required.
- **Human variability:** outcomes depend on culture, training, incentives; causal designs must control confounders.
- **Ethical constraints:** strongest causal tests may be limited by ethics and consent.

### 17.3 Predictions (testable)

If pre-singularity is real as defined here, we predict:

- change points in outcome time series aligned with MVF deployments,
- rising explanatory power of  $\mathbf{E}$  interventions over long-horizon behavior,
- percolation-like transitions from fragmented to coherent  $G_E$ ,
- new fragility modes: correlated failures, adversarial exploit surfaces, objective monocultures.

## 18 Conclusion

This manuscript reframes singularity-like transitions as *environmental agency emergence* rather than isolated model capability thresholds. The pre-singularity, operationally, is the regime in which the sub-environment becomes coherent, low-latency, stateful, and causally dominant enough to architect human trajectories by shaping constraints and defaults continuously. We provide a mechanism-first formalization, conservative diagnostics, and test protocols intended to be credible under skeptical review. The engineering question becomes:

*How do we design computational environments that become more capable while staying stable, auditable, overridable, decentralized, and privacy-respecting?*

## A Appendix A: Formal Field Intelligence Functionals

**Definition 10** (Task class). *Let  $\Pi$  be a class of tasks with constraints (energy, time, safety). For each  $\pi \in \Pi$ , define a performance functional:*

$$\mathcal{P}_\pi(\mathbf{H}, \mathbf{A}, \mathbf{E}) \in \mathbb{R}. \quad (54)$$

**Definition 11** (Field intelligence). *Define:*

$$I_{field} \triangleq \mathbb{E}_{\pi \sim \Pi}[\mathcal{P}_\pi(\mathbf{H}, \mathbf{A}, \mathbf{E})], \quad (55)$$

*and baseline local intelligence with passive environment  $\mathbf{E}_0$ :*

$$I_{local} \triangleq \mathbb{E}_{\pi \sim \Pi}[\mathcal{P}_\pi(\mathbf{H}, \mathbf{A}, \mathbf{E}_0)]. \quad (56)$$

**Proposition 2** (Proxy amplification). *If  $\mathbf{E}$  reduces effective latency and increases observability under fixed energy bounds, then there exists  $\Pi' \subseteq \Pi$  such that:*

$$I_{field}(\Pi') > I_{local}(\Pi'). \quad (57)$$

## B Appendix B: Minimal Data Schema for Measuring PSI

A practical MVF test logs:

- synchronized timestamps (NTP/PTP where possible),
- sensor summaries  $y_E(t)$  (privacy-preserving aggregates),
- state estimates  $\hat{s}_E(t)$  and confidence,
- actions  $u_E(t)$  applied actuation + bounds/projection,
- graph snapshots for  $G_E(t)$  to compute  $\kappa_E$ ,
- outcomes  $Y(t)$  (time, cost, incidents),
- interventions  $Z(t)$  for causal tests,
- overrides + resolution times,
- audit traces (objective terms, constraint activations).

## C Appendix C: Field Layers Diagram Template

Physical world (geometry, matter, motion)

Sensing field (multi-modal observation)

Inference field (state estimation, prediction)

Actuation field (routing, cues, control)

Memory/energy field (place-memory, routing, harvesting)

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